

LARGE SCALE DATA ANALYTICS

GEO_SPARK

SCALABILITY EVALUATION OF K-MEANS CLUSTERING ON 2 MILLION 128-DIMENSIONAL MAP EMBEDDINGS, COMPARING FIVE IMPLEMENTATIONS, USING PROGRESSIVE SIZE-UP TESTS, AND INCORPORATING INTERPRETABILITY THROUGH KNN.

PROJECT OVERVIEW

This project presents a comprehensive scalability study of distributed K-means clustering on large-scale geospatial datasets using Apache Spark. It processes 128-dimensional **Sentinel-2** satellite image embeddings from the **GeoTessera** dataset to cluster land cover types. The study benchmarks five different implementations using a **Size-Up** methodology on approximately 2 million data points. Beyond performance analysis, the project bridges the gap between unsupervised clustering and **interpretability** by integrating supervised K-Nearest Neighbors (KNN) classification, producing georeferenced visualizations suitable for GIS applications.

OBJECTIVE

The primary research goal is to answer three questions:

- how distinct K-means **code variations scale** as data volumes increase,
- what **trade-offs** exist between implementation complexity and execution performance
- how to effectively map unsupervised clusters to **human-readable** categories.

The study seeks to identify the optimal **balance** between code abstraction and computational efficiency while validating the utility of hybrid unsupervised-supervised workflows.

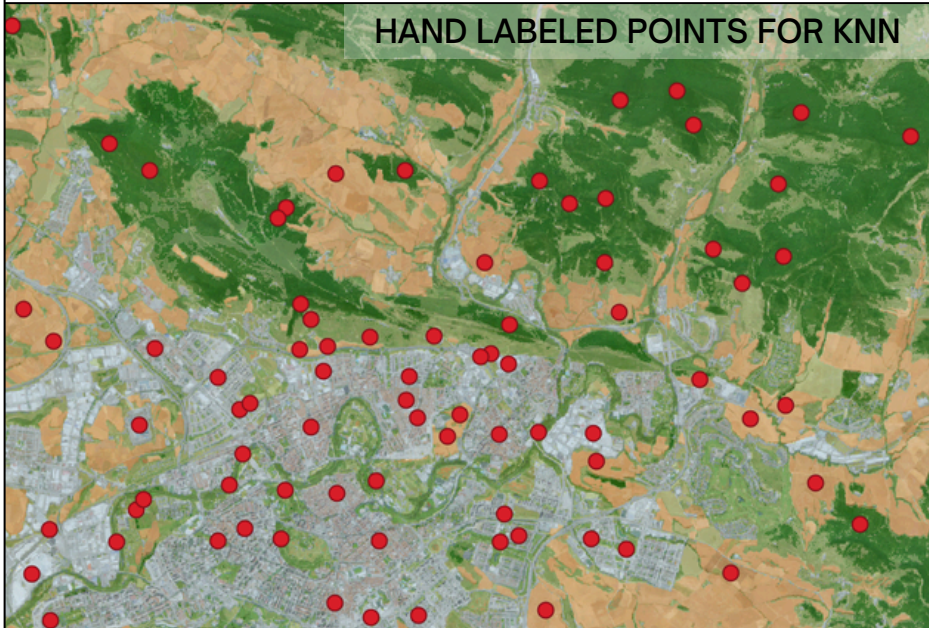


DATASET

The source material comes from the GeoTessera project, which in our case is refocused on the Pamplona region of Spain. Technical specifications include:

- Source:** ~2 million map tile embeddings (128 dimensions) derived from Sentinel-2 satellite imagery at a **10-meter** resolution
- Domain:** Geospatial/cartographic feature representations
- Format:** Parquet (optimized columnar storage)
- Size:** Up to **1GB** in our tests, but scalability is **virtually infinite** (expanding input map section). Sufficient to test single-machine vs. cluster trade-offs

The secondary control dataset consists of 80 **manually labeled** ground-truth points (20 each for Vegetation, Forest, Urban, and Agricultural classes) stored as GeoJSON.



DATA PIPELINE

The processing workflow moves from raw storage to visualization. It begins by ingesting raw embedding Parquet files and hand-labeled point **geojson** ensuring correct Coordinate Reference System (EPSG:4326 to EPSG:32630). After that, a KD-Tree spatial index is built to associate labeled points with nearest-neighbor embeddings efficiently. Following this preparation, the pipeline executes KNN model training and prediction, finally exporting the results as spatially referenced **GeoTIFF mosaics**.

METHODOLOGY & ANALYSIS

The analytic approach correlates quantitative benchmarks with qualitative geospatial validation. The study assesses the **linearity**, or lack thereof, of the Size-Up curves to identify bottlenecks, such as memory limitations of the **local** implementation versus the high overhead of some **global** solutions. Simultaneously, clustering quality is verified by cross-referencing K-means centroids with KNN predictions in reduced dimensionality (**PCA**) space and visually inspecting the thematic maps in **QGIS** for spatial coherence.

IMPLEMENTATIONS EVALUATED

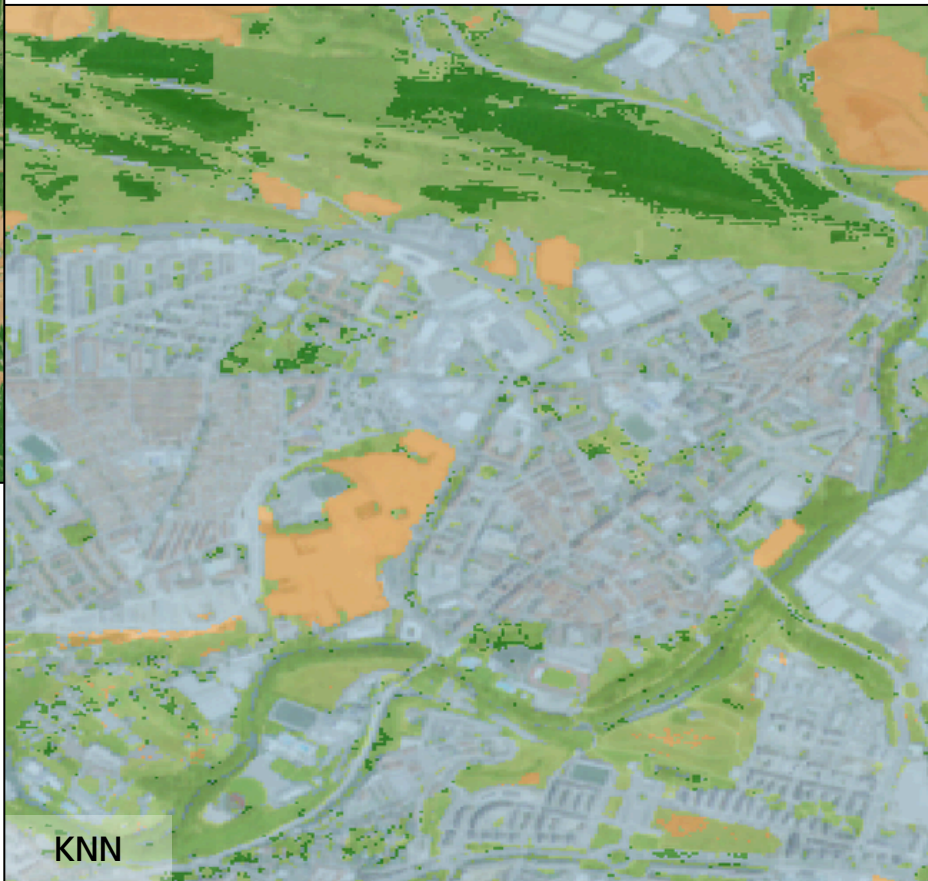
Five distinct algorithmic approaches were implemented to isolate performance variables. These range from a baseline single-node **Local NumPy** script to specific Spark variations:

- the standard **MLlib** library
- a low-level **Custom RDD** MapReduce implementation
- a **DataFrame** approach using Python **UDFs**
- an Optimized **DataFrame** version utilizing native **Spark SQL** functions.

This selection compares **standard** library efficiency against **custom** low-level and high-level coding patterns.

TESTING STRATEGY

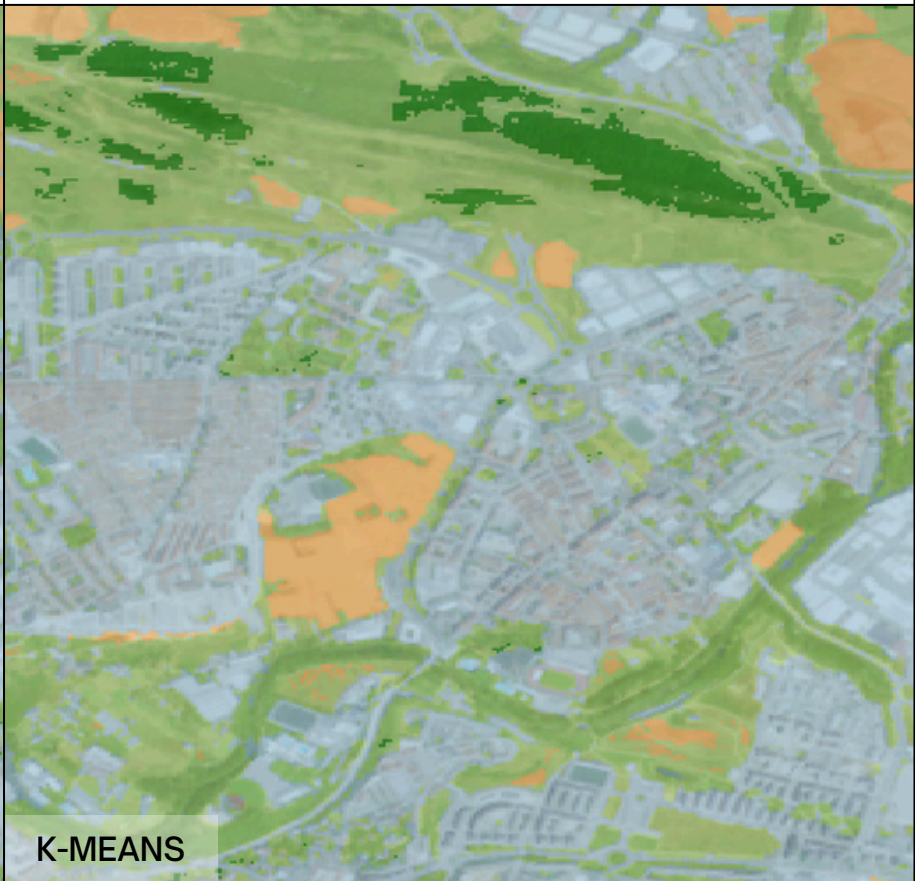
To measure scalability, a **Size-Up** experimental design was employed. Each implementation was executed against dataset samples at 10%-25%-50%-75%-100% of the total volume. The primary metrics recorded were wall-clock training time, speedup ratios relative to the baseline, and convergence accuracy **WSSSE** (Within-Set Sum of Squared Errors). Experiments used fixed random seeds and same resource availability (16 GB RAM and all machine cores) to ensure reproducibility and fair comparison.



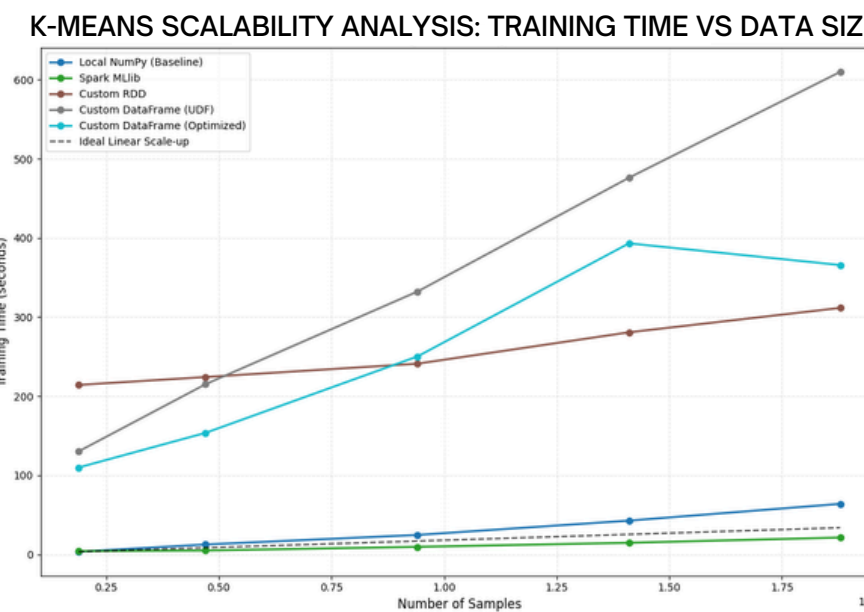
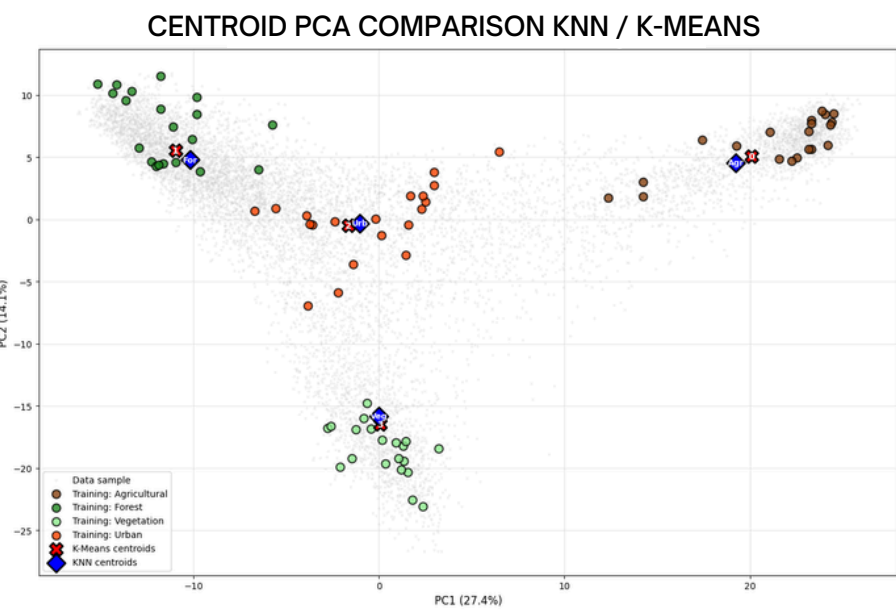
Clasification map overlay on top of IDENA ORTOFOTO WMS Satellite imagery
| Grey: Urban | Yellow: Agriculture | Dark-Green: Forest | Light-Green: Vegetation

STUDYING THE RESULTS

Different approaches were taken to interpret the predictions obtained. Aside from simply overlaying the class mask on the original Sentinel-2 input data, we used IDENA orthophoto satellite imagery, which allowed studying the underlying info in greater detail, as the pixel resolution is 20 cm instead of 10 m. On a more analytical level, the centroid distribution of the classes was studied via a PCA plot of the embeddings.



RESULTS



CONCLUSION AND FUTURE WORK

The study concludes that Spark MLlib provides the superior performance of **below-linear** scalability and ease of use, while custom RDD and DataFrame implementations offer educational insight but incur significant **overhead**. Additionally, although in these experiments the local solution presents **lower training times**, its main drawback was the **high memory demand**, which made it even unfeasible to run the experiment on less powerful machines. The hybrid KNN-Kmeans workflow proved effective for generating **interpretable maps**, backed up by the clusters' centroid proximity. Future research could extend this work to multi-node cluster environments to test network overhead, explore GPU acceleration, and evaluate density-based algorithms like DBSCAN.

NOTES

The integration of **distributed computing** with **geospatial machine** learning represents a crucial capability for modern Earth observation science as satellite data volumes continue to grow **exponentially**.

References:

[1]M. Drusch et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," Remote sensing of Environment, vol. 120, 2012.

[2]J. MacQueen and others, "Some methods for classification and analysis of multivariate observations," Proceedings of the fifth Berkeley symposium on mathematical statistics and probability, vol. 1, no. 14, 1967.

[3]X. Meng et al., "MLlib: Machine learning in apache spark," The Journal of Machine Learning Research, vol. 17, no. 1, 2016.

[4]J. Jakubik et al., "Foundation models for generalist geospatial artificial intelligence," arXiv preprint arXiv:2310.18660, 2023.

[5]J. L. Bentley, "Multidimensional binary search trees used for associative searching," Communications of the ACM, vol. 18, no. 9,